

1. Screenshots (label each clearly)

- Pods/services running in tenant-a and tenant-b (Task 1).

```
(melon@melon)-[~]  
$ kubectl create namespace tenant-a  
kubectl create namespace tenant-b  
  
# Deploy a simple web server for each tenant  
kubectl -n tenant-a create deployment web --image=nginx  
kubectl -n tenant-b create deployment web --image=nginx  
kubectl -n tenant-a expose deployment web --port=80  
kubectl -n tenant-b expose deployment web --port=80  
kubectl get pods,svc -n tenant-a  
  
namespace/tenant-a created  
namespace/tenant-b created  
deployment.apps/web created  
deployment.apps/web created  
service/web exposed  
service/web exposed  


| NAME                     | READY | STATUS            | RESTARTS | AGE |
|--------------------------|-------|-------------------|----------|-----|
| pod/web-7c56dcdb9b-wnrfg | 0/1   | ContainerCreating | 0        | 1s  |

  


| NAME        | TYPE      | CLUSTER-IP    | EXTERNAL-IP | PORT(S) | AGE |
|-------------|-----------|---------------|-------------|---------|-----|
| service/web | ClusterIP | 10.96.181.248 | <none>      | 80/TCP  | 1s  |


```

- The **before** probe result: HTTP 200 across tenants (Task 2).

```
(melon@melon)-[~]  
$ # From tenant-a, curl tenant-b (replace <B_IP>)  
kubectl -n tenant-a run probe --rm -it --image=curlimages/curl --restart=Never \\\n-- curl -s -m 5 http://10.96.173.224 -o /dev/null -w 'HTTP %{http_code}\\n'  
  
HTTP 200  
pod "probe" deleted
```

- The **after** probe result: timeout once the NetworkPolicy is applied (Task 4).

```
(melon@melon)-[~]
$ # From tenant-a, curl tenant-b (replace <B_IP>)
kubectl -n tenant-a run probe --rm -it --image=curlimages/curl --restart=Never \
-- curl -s -m 5 http://10.96.173.224 -o /dev/null -w 'HTTP %{http_code}\n'

Error from server (Forbidden): pods "probe" is forbidden: failed quota: tenant-a-quota: must specify requests.cpu for: probe; requests.memory for: probe
```

- The two `auth can-i` results proving secret isolation (Task 5).

```
(melon@melon)-[~]
$ # A service account scoped to tenant-a only
kubectl -n tenant-a create serviceaccount app-a
kubectl -n tenant-a create role reader --verb=get --resource=secrets
kubectl -n tenant-a create rolebinding rb --role=reader --serviceaccount=tenant-a:app-a

SA=system:serviceaccount:tenant-a:app-a
kubectl auth can-i get secrets -n tenant-a --as=$SA # expect: yes
kubectl auth can-i get secrets -n tenant-b --as=$SA # expect: no

serviceaccount/app-a created
role.rbac.authorization.k8s.io/reader created
rolebinding.rbac.authorization.k8s.io/rb created
yes
no
```

- The remanence scan and the secure-wipe output (Task 6).

```
(melon@melon)-[~]
$ # Secure wipe: overwrite before delete (shred)
docker run --rm -v ccse-vol:/data alpine sh -c \
'echo SENSITIVE > /data/phi2.txt; sync; \
dd if=/dev/zero of=/data/phi2.txt bs=1k count=1 conv=notrunc; rm /data/phi2.txt; echo wiped'

1+0 records in
1+0 records out
1024 bytes (1.0KB) copied, 0.000022 seconds, 44.4MB/s
wiped
```